Aaradhaya Sutrave

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EDUCATION

Texas A&M University (TAMU) Expected Dec. 2026

- Bachelor of Science in Mechanical Engineering | College Station, TX
- Relevant coursework: Heat Transfer, Dynamic Systems and Controls, Thermodynamics, Fluid Mechanics, Materials and Manufacturing in Design, Computer-Aided Engineering

TECHNICAL SKILLS

Programming & Data Acquisition: MATLAB, Python, LabVIEW, Arduino Programming, Excel
Optics & Instrumentation: 633 nm laser, spatial light modulator (SLM), lens/optical path alignment, lock-in amplifier, oscilloscope, function generator, NSOM probe, circuit simulation
Design & Analysis: SolidWorks, ANSYS, Fusion 360, Orca Slicer, 3D printing

EXPERIENCE

Undergraduate Research Assistant – Sy Bor Wen, Department of Mechanical Engineering May 2026 - Present

- **Holographic Vision (in progress):** Generating 2D holograms with a 633 nm laser and spatial light modulator (SLM), building a multi-lens optical path (25, 100, 75 and 200 mm lenses) as groundwork for 3D holograms to help visualize fine detail in surgery and artificial human parts
- **Microheater Optics (completed):** Developed a LabVIEW VI that synchronized a servo-driven lens scan with lock-in amplifier buffered capture and exported combined scan data to CSV
- Demodulated lock-in data against the reference signal in MATLAB to produce FFTs of the microheater's optical response, using a function generator, oscilloscope and NSOM probe to generate and measure signals
- Contributing to an ongoing research paper on Near Field Radiation

Undergraduate Research Assistant – MIDL Research Lab Feb. 2026 - Present

- Developed MATLAB code to generate Voronoi diagrams and half-edge mesh data structures that replicate finite element analysis meshes
- Designed precision acoustic test apparatus (Kundt tube) parts in SolidWorks, including PVC tube flanges, speaker reducers and measurement ports, then converted designs to G-code for 3D printing
- Contributing to an ongoing research manuscript on noise absorption of weave structures

ENGINEERING PROJECTS

CNC Machining and CAM Project – Fusion 360 Apr. 2026

- Created a CAD model, generated CAM toolpaths and tool selection in Fusion 360, and machined a plastic part on a CNC mill

Structural Design & FEA on Playground Equipment Apr. 2026

- Redesigned a multi-person playground slide in galvanized steel instead of plastic for lifespan, durability, and sustainability and created a bill of materials for manufacturing
- Analyzed bending stress, shear force, deformation and factor of safety in ANSYS, then revised the design for safety

Backseat Minder – Sensor, Circuit, & Alert-System Design Jan. 2026

- Created circuit diagram and simulation for a self-charging monitoring concept with occupant, temperature, and humidity detection
- Designed system functionality for alerts through a cellular module and supported CAD work for PCB integration considerations

Aluminum Hammer Fabrication: Multi Process Manufacturing Nov. 2025

- Fabricated an aluminum hammer using manual milling, lathe turning and precision sanding in 4 - 5 weeks

Aerodynamic Car Design & Wind Tunnel Testing – SolidWorks, Flow Simulation Nov. 2025

- Led a four-person team to design and test a low-drag car body using SolidWorks and Flow Simulation
- Validated the design through wind-tunnel testing and achieved the lowest drag coefficient of the 12 teams

Stepped Shaft FEA/Discontinuity Effects Nov. 2025 - Present

- Conducted FEA of a load-bearing stepped shaft under different loading and constraints in SolidWorks, evaluating bending, von Mises, and shear stress to assess the effects of geometric discontinuities
- Developed numerical simulations to validate FEA results and identify stress concentrations and potential failure spots

Mars Rover Simulation – Python Modeling & Optimization Oct. 2024

- With a team, built a Python simulation of descent, parachute deployment and driving over randomized Mars terrain, parameterizing rover mass, materials and DC motor behavior
- Optimized design parameters for speed across any terrain; final design covered 1,000 m in about 2,843 s with a 161.5 W peak motor power